WEEK 6 - Product Review And Rating Management System

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Scenario:

The Product Review and Rating Management System is a database application used to collect and manage customer feedback about purchased products. It allows customers to share their opinions and rate products based on quality, design, durability, and overall satisfaction.

This system connects customer information with product details and helps identify popular products and areas that require improvement. It also helps businesses understand customer preferences and improve product quality.

Review and Rating Table Creation

A Review table is used to store customer feedback and product ratings. It contains details such as Review ID, Customer ID, Product ID, Rating, Review Comment, and Review Date.

A rating constraint ensures that customers can provide ratings only between 1 and 5, where 1 represents very poor satisfaction and 5 represents excellent satisfaction.

```
SQL> CREATE TABLE Review (
  2      Review_ID VARCHAR2(5) PRIMARY KEY,
  3      Customer_ID VARCHAR2(5),
  4      Product_ID VARCHAR2(7),
  5      Rating NUMBER(1) CHECK (Rating BETWEEN 1 AND 5),
  6      Review_Comment VARCHAR2(200),
  7      Review_Date DATE,
  8      FOREIGN KEY (Customer_ID) REFERENCES Customer(Customer_ID),
  9      FOREIGN KEY (Product_ID) REFERENCES Product(Product_ID)
 10  );

Table created.
```

```
SQL> desc Review;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 REVIEW_ID                                 NOT NULL VARCHAR2(5)
 CUSTOMER_ID                                        VARCHAR2(5)
 PRODUCT_ID                                         VARCHAR2(7)
 RATING                                             NUMBER(1)
 REVIEW_COMMENT                                     VARCHAR2(200)
 REVIEW_DATE                                        DATE
```

Inserting values

```
SQL> INSERT INTO Review VALUES
  2  ('R001', 'C001', 'P000001', 5, 'Good quality case', '27-SEP-26');

1 row created.

SQL> INSERT INTO Review VALUES
  2  ('R002', 'C001', 'P000002', 4, 'Good silicone case', '27-SEP-26');

1 row created.

SQL> INSERT INTO Review VALUES
  2  ('R003', 'C002', 'P000003', 5, 'Very clear and attractive', '28-SEP-26');

1 row created.

SQL>
SQL> INSERT INTO Review VALUES
  2  ('R004', 'C003', 'P000004', 4, 'Nice printed design', '29-SEP-26');

1 row created.

SQL>
SQL> INSERT INTO Review VALUES
  2  ('R005', 'C001', 'P000005', 5, 'Excellent designer case', '30-SEP-26');

1 row created.

SQL>
SQL> INSERT INTO Review VALUES
  2  ('R006', 'C004', 'P000001', 4, 'Strong and durable case', '30-SEP-26');

1 row created.

SQL>
SQL> INSERT INTO Review VALUES
  2  ('R007', 'C005', 'P000002', 5, 'Soft and comfortable', '01-OCT-26');

1 row created.
```

```
SQL> INSERT INTO Review VALUES
  2  ('R009', 'C003', 'P000004', 5, 'Beautiful design', '01-OCT-26');

1 row created.

SQL>
SQL> INSERT INTO Review VALUES
  2  ('R010', 'C005', 'P000005', 4, 'Looks premium', '02-OCT-26');

1 row created.

SQL>
SQL> INSERT INTO Review VALUES
  2  ('R011', 'C001', 'P000006', 5, 'Good photo case', '02-OCT-26');

1 row created.

SQL>
SQL> INSERT INTO Review VALUES
  2  ('R012', 'C002', 'P000007', 5, 'Very strong case', '03-OCT-26');

1 row created.

SQL>
SQL> INSERT INTO Review VALUES
  2  ('R013', 'C003', 'P000008', 4, 'Simple and useful', '03-OCT-26');

1 row created.

SQL>
SQL> INSERT INTO Review VALUES
  2  ('R014', 'C004', 'P000009', 5, 'Nice black case', '04-OCT-26');

1 row created.

SQL>
SQL> INSERT INTO Review VALUES
  2  ('R015', 'C005', 'P000010', 4, 'Slim and good quality', '04-OCT-26');

1 row created.
```

Updating reviews: Modify review comments when necessary.

Deleting reviews: Remove inappropriate, invalid, or unwanted reviews.

```
SQL> UPDATE Review
  2  SET Review_Comment = 'Very good quality and design'
  3  WHERE Review_ID = 'R001';

1 row updated.

SQL> DELETE FROM Review
  2  WHERE Review_ID = 'R004';

1 row deleted.
```

Retrieving All Product Review Details

Retrieving review details helps businesses understand customer opinions about their products. The database can display all reviews associated with a particular product and show customer names along with their comments and rating.

```
SQL> select * from Review;

REVIE CUSTO PRODUCT     RATING
----- ----- ------- ----------
REVIEW_COMMENT
--------------------------------------------------------------
REVIEW_DA
---------
R001  C001  P000001          5
Very good quality and design
27-SEP-26

R002  C001  P000002          4
Good silicone case
27-SEP-26

REVIE CUSTO PRODUCT     RATING
----- ----- ------- ----------
REVIEW_COMMENT
--------------------------------------------------------------
REVIEW_DA
---------

R003  C002  P000003          5
Very clear and attractive
28-SEP-26

R005  C001  P000005          5
Excellent designer case
```

```
REVIE CUSTO PRODUCT     RATING
----- ----- ------- ----------
REVIEW_COMMENT
--------------------------------------------------------------------
REVIEW_DA
---------
30-SEP-26

R006  C004  P000001          4
Strong and durable case
30-SEP-26

R007  C005  P000002          5

REVIE CUSTO PRODUCT     RATING
----- ----- ------- ----------
REVIEW_COMMENT
--------------------------------------------------------------------
REVIEW_DA
---------
Soft and comfortable
01-OCT-26

R008  C002  P000003          4
Good product quality
01-OCT-26


REVIE CUSTO PRODUCT     RATING
----- ----- ------- ----------
REVIEW_COMMENT
--------------------------------------------------------------------
REVIEW_DA
---------
R009  C003  P000004          5
Beautiful design
01-OCT-26
```

```
R010  C005  P000005          4
Looks premium
02-OCT-26

REVIE CUSTO PRODUCT     RATING
----- ----- ------- ----------
REVIEW_COMMENT
--------------------------------------------------------------------
REVIEW_DA
---------

R011  C001  P000006          5
Good photo case
02-OCT-26

R012  C002  P000007          5
Very strong case

REVIE CUSTO PRODUCT     RATING
----- ----- ------- ----------
REVIEW_COMMENT
--------------------------------------------------------------------
REVIEW_DA
---------
03-OCT-26

R013  C003  P000008          4
Simple and useful
03-OCT-26
```

```
REVIE CUSTO PRODUCT     RATING
----- ----- ------- ----------
REVIEW_COMMENT
----------------------------------------
REVIEW_DA
---------
Nice black case
04-OCT-26

R015  C005  P000010          4
Slim and good quality
04-OCT-26


14 rows selected.
```

- **Retrieve Product Review Details**

1. Display all reviews for a product

```
SQL> SELECT *
  2  FROM Review
  3  WHERE Product_ID = 'P000001';

REVIE CUSTO PRODUCT     RATING
----- ----- ------- ----------
REVIEW_COMMENT
--------------------------------------------------
REVIEW_DA
---------
R001  C001  P000001          5
Very good quality and design
27-SEP-26

R006  C004  P000001          4
Strong and durable case
30-SEP-26
```

2. Display customer name with their reviews

```
SQL> SELECT C.Customer_Name,
  2         R.Product_ID,
  3         R.Rating,
  4         R.Review_Comment,
  5         R.Review_Date
  6  FROM Customer C
  7  JOIN Review R
  8  ON C.Customer_ID = R.Customer_ID;

CUSTOMER_NAME                  PRODUCT     RATING
------------------------------ ------- ----------
REVIEW_COMMENT
--------------------------------------------------------------
REVIEW_DA
---------
Arun Kumar                     P000001          5
Very good quality and design
27-SEP-26

Arun Kumar                     P000002          4
Good silicone case
27-SEP-26

CUSTOMER_NAME                  PRODUCT     RATING
------------------------------ ------- ----------
REVIEW_COMMENT
--------------------------------------------------------------
REVIEW_DA
---------

Aakash                         P000003          5
Very clear and attractive
28-SEP-26

Arun Kumar                     P000005          5
Excellent designer case
```

```
CUSTOMER_NAME                  PRODUCT     RATING
------------------------------ ------- ----------
REVIEW_COMMENT
--------------------------------------------------------------
REVIEW_DA
---------
30-SEP-26

Divya Kumar                    P000001          4
Strong and durable case
30-SEP-26

Karthik S                      P000002          5

CUSTOMER_NAME                  PRODUCT     RATING
------------------------------ ------- ----------
REVIEW_COMMENT
--------------------------------------------------------------
REVIEW_DA
---------
Soft and comfortable
01-OCT-26

Aakash                         P000003          4
Good product quality
01-OCT-26


CUSTOMER_NAME                  PRODUCT     RATING
------------------------------ ------- ----------
REVIEW_COMMENT
--------------------------------------------------------------
REVIEW_DA
---------
Rahul Raj                      P000004          5
Beautiful design
01-OCT-26
```

```
Karthik S                      P000005          4
Looks premium
02-OCT-26

CUSTOMER_NAME                  PRODUCT     RATING
------------------------------ ------- ----------
REVIEW_COMMENT
--------------------------------------------------------------------
REVIEW_DA
---------

Arun Kumar                     P000006          5
Good photo case
02-OCT-26

Aakash                         P000007          5
Very strong case

CUSTOMER_NAME                  PRODUCT     RATING
------------------------------ ------- ----------
REVIEW_COMMENT
--------------------------------------------------------------------
REVIEW_DA
---------
03-OCT-26

Rahul Raj                      P000008          4
Simple and useful
03-OCT-26

Divya Kumar                    P000009          5
```

```
CUSTOMER_NAME                  PRODUCT     RATING
------------------------------ ------- ----------
REVIEW_COMMENT
--------------------------------------------------------------------
REVIEW_DA
---------
Nice black case
04-OCT-26

Karthik S                      P000010          4
Slim and good quality
04-OCT-26


14 rows selected.
```

3. Find products having maximum reviews

```
SQL> SELECT P.PRODUCT_ID,
  2         P.PRODUCT_NAME,
  3         COUNT(R.REVIEW_ID) AS Number_Of_Reviews
  4  FROM Product P
  5  JOIN Review R
  6  ON P.PRODUCT_ID = R.PRODUCT_ID
  7  GROUP BY P.PRODUCT_ID, P.PRODUCT_NAME
  8  ORDER BY Number_Of_Reviews DESC;

PRODUCT
-------
PRODUCT_NAME
------------------------------------------------------------
NUMBER_OF_REVIEWS
-----------------
P000001
Clear Shock Case
                2

P000003
Crystal Clear Case
                2

PRODUCT
-------
PRODUCT_NAME
------------------------------------------------------------
NUMBER_OF_REVIEWS
-----------------

P000005
Luxury Designer Case
                2

P000002
Soft Silicone Case
```

```
PRODUCT
-------
PRODUCT_NAME
------------------------------------------------------------
NUMBER_OF_REVIEWS
-----------------
                2

P000007
Heavy Duty Case
                1

P000009

PRODUCT
-------
PRODUCT_NAME
------------------------------------------------------------
NUMBER_OF_REVIEWS
-----------------
Classic Black Case
                1

P000006
Personal Photo Case
                1


PRODUCT
-------
PRODUCT_NAME
------------------------------------------------------------
NUMBER_OF_REVIEWS
-----------------
P000010
Slim Basic Case
                1
```

```
P000008
Matte Basic Case
               1

PRODUCT
-------
PRODUCT_NAME
------------------------------------------------------------
NUMBER_OF_REVIEWS
-----------------

P000004
Floral Printed Case
               1


10 rows selected.
```

4. Display recent customer feedback

```
SQL> SELECT C.Customer_Name,
  2         R.Product_ID,
  3         R.Review_Comment,
  4         R.Review_Date
  5  FROM Customer C
  6  JOIN Review R
  7  ON C.Customer_ID = R.Customer_ID
  8  ORDER BY R.Review_Date DESC;

CUSTOMER_NAME                  PRODUCT
------------------------------ -------
REVIEW_COMMENT
------------------------------------------------------------
REVIEW_DA
---------
Karthik S                      P000010
Slim and good quality
04-OCT-26

Divya Kumar                    P000009
Nice black case
04-OCT-26

CUSTOMER_NAME                  PRODUCT
------------------------------ -------
REVIEW_COMMENT
------------------------------------------------------------
REVIEW_DA
---------

Rahul Raj                      P000008
Simple and useful
03-OCT-26
```

```
Aakash                         P000007
Very strong case

CUSTOMER_NAME                  PRODUCT
------------------------------ -------
REVIEW_COMMENT
--------------------------------------------------------------------
REVIEW_DA
---------
03-OCT-26

Arun Kumar                     P000006
Good photo case
02-OCT-26

Karthik S                      P000005

CUSTOMER_NAME                  PRODUCT
------------------------------ -------
REVIEW_COMMENT
--------------------------------------------------------------------
REVIEW_DA
---------
Looks premium
02-OCT-26

Rahul Raj                      P000004
Beautiful design
01-OCT-26
```

```
Arun Kumar                       P000005
Excellent designer case
30-SEP-26

Divya Kumar                      P000001
Strong and durable case

CUSTOMER_NAME                  PRODUCT
------------------------------ --------
REVIEW_COMMENT
----------------------------------------------------------------
REVIEW_DA
---------
30-SEP-26

Aakash                           P000003
Very clear and attractive
28-SEP-26

Arun Kumar                       P000002

CUSTOMER_NAME                  PRODUCT
------------------------------ -------
REVIEW_COMMENT
----------------------------------------------------------------
REVIEW_DA
---------
Good silicone case
27-SEP-26

Arun Kumar                       P000001
Very good quality and design
27-SEP-26


14 rows selected.
```

5. Retrieve reviews with ratings above 4

```
SQL> SELECT *
  2  FROM Review
  3  WHERE Rating > 4;

REVIE CUSTO PRODUCT     RATING
----- ----- ------- ----------
REVIEW_COMMENT
--------------------------------------------------------------------------------
REVIEW_DA
---------
R001  C001  P000001          5
Very good quality and design
27-SEP-26

R003  C002  P000003          5
Very clear and attractive
28-SEP-26

REVIE CUSTO PRODUCT     RATING
----- ----- ------- ----------
REVIEW_COMMENT
--------------------------------------------------------------------------------
REVIEW_DA
---------

R005  C001  P000005          5
Excellent designer case
30-SEP-26

R007  C005  P000002          5
Soft and comfortable
```

```
REVIE CUSTO PRODUCT     RATING
----- ----- ------- ----------
REVIEW_COMMENT
--------------------------------------------------------------------------------
REVIEW_DA
---------
Good photo case
02-OCT-26

R012  C002  P000007          5
Very strong case
03-OCT-26


REVIE CUSTO PRODUCT     RATING
----- ----- ------- ----------
REVIEW_COMMENT
--------------------------------------------------------------------------------
REVIEW_DA
---------
R014  C004  P000009          5
Nice black case
04-OCT-26


8 rows selected.
```

- ❖ **Calculate Average Product Ratings**

1. Average rating for each product

```
SQL> SELECT Product_ID,
  2         AVG(Rating) AS Average_Rating
  3  FROM Review
  4  GROUP BY Product_ID;

PRODUCT AVERAGE_RATING
------- --------------
P000003            4.5
P000009              5
P000002            4.5
P000008              4
P000005            4.5
P000006              5
P000001            4.5
P000010              4
P000004              5
P000007              5

10 rows selected.
```

2. **Count number of reviews per product**

```
SQL> SELECT Product_ID,
  2         COUNT(*) AS Number_Of_Reviews
  3  FROM Review
  4  GROUP BY Product_ID;

PRODUCT NUMBER_OF_REVIEWS
------- -----------------
P000003                 2
P000009                 1
P000002                 2
P000008                 1
P000005                 2
P000006                 1
P000001                 2
P000010                 1
P000004                 1
P000007                 1

10 rows selected.
```

3. Find highest-rated products

```
SQL> SELECT Product_ID,
  2        AVG(Rating) AS Average_Rating
  3  FROM Review
  4  GROUP BY Product_ID
  5  ORDER BY Average_Rating DESC;

PRODUCT AVERAGE_RATING
------- --------------
P000009              5
P000006              5
P000007              5
P000004              5
P000001            4.5
P000002            4.5
P000003            4.5
P000005            4.5
P000008              4
P000010              4

10 rows selected.
```

4. Products with average rating above 4

```
SQL> SELECT Product_ID,
  2         AVG(Rating) AS Average_Rating
  3  FROM Review
  4  GROUP BY Product_ID
  5  HAVING AVG(Rating) > 4;

PRODUCT AVERAGE_RATING
------- --------------
P000003            4.5
P000009              5
P000002            4.5
P000005            4.5
P000006              5
P000001            4.5
P000004              5
P000007              5

8 rows selected.
```

Reports

Report 1: Product Rating Analysis

The Product Rating Analysis report provides an overview of product performance. It includes product names, the number of reviews received, and the average rating of each product.

This report helps compare products and identify those that receive positive customer feedback.

```
SQL> SELECT P.Product_Name,
  2         COUNT(R.Review_ID) AS Number_Of_Reviews,
  3         ROUND(AVG(R.Rating), 2) AS Average_Rating
  4  FROM Product P
  5  LEFT JOIN Review R
  6  ON P.Product_ID = R.Product_ID
  7  GROUP BY P.Product_ID, P.Product_Name
  8  ORDER BY Average_Rating DESC;

PRODUCT_NAME
--------------------------------------------------------
NUMBER_OF_REVIEWS AVERAGE_RATING
----------------- --------------
Floral Print Case
                0

Everyday Basic Case
                0

Collage Photo Case
                0


PRODUCT_NAME
--------------------------------------------------------
NUMBER_OF_REVIEWS AVERAGE_RATING
----------------- --------------
Abstract Art Case
                0

Military Grade Case
                0
```

```
Floral Printed Case
                1              5

Personal Photo Case
                1              5


PRODUCT_NAME
--------------------------------------------------
NUMBER_OF_REVIEWS AVERAGE_RATING
----------------- --------------
Soft Silicone Case
                2            4.5

Luxury Designer Case
                2            4.5

Clear Shock Case
                2            4.5


PRODUCT_NAME
--------------------------------------------------
NUMBER_OF_REVIEWS AVERAGE_RATING
----------------- --------------
Crystal Clear Case
                2            4.5

Slim Basic Case
                1              4

Matte Basic Case
                1              4


30 rows selected.
```

Report 2: Customer Feedback Analysis

The Customer Feedback Analysis report examines customer opinions and review activity. It identifies the most-reviewed products, highly rated products, and products that may require improvement.

This report helps businesses understand customer expectations and make suitable decisions to improve their products.

- **Most reviewed products:**

```
SQL> SELECT P.Product_Name,
  2         COUNT(R.Review_ID) AS Number_Of_Reviews
  3  FROM Product P
  4  JOIN Review R
  5  ON P.Product_ID = R.Product_ID
  6  GROUP BY P.Product_ID, P.Product_Name
  7  ORDER BY Number_Of_Reviews DESC;

PRODUCT_NAME
--------------------------------------------------------
NUMBER_OF_REVIEWS
-----------------
Clear Shock Case
                2

Crystal Clear Case
                2

Luxury Designer Case
                2


PRODUCT_NAME
--------------------------------------------------------
NUMBER_OF_REVIEWS
-----------------
Soft Silicone Case
                2

Heavy Duty Case
                1

Classic Black Case
                1
```

```
PRODUCT_NAME
------------------------------------------------
NUMBER_OF_REVIEWS
-----------------
Personal Photo Case
                 1

Slim Basic Case
                 1

Matte Basic Case
                 1


PRODUCT_NAME
------------------------------------------------
NUMBER_OF_REVIEWS
-----------------
Floral Printed Case
                 1


10 rows selected.
```

- **Highly rated products**:

```
SQL> SELECT P.Product_Name,
  2         ROUND(AVG(R.Rating), 2) AS Average_Rating
  3  FROM Product P
  4  JOIN Review R
  5  ON P.Product_ID = R.Product_ID
  6  GROUP BY P.Product_ID, P.Product_Name
  7  HAVING AVG(R.Rating) >= 4;

PRODUCT_NAME
--------------------------------------------------------------------
AVERAGE_RATING
--------------
Soft Silicone Case
           4.5

Luxury Designer Case
           4.5

Matte Basic Case
            4


PRODUCT_NAME
--------------------------------------------------------------------
AVERAGE_RATING
--------------
Slim Basic Case
            4

Floral Printed Case
            5

Personal Photo Case
            5
```

```
PRODUCT_NAME
--------------------------------------------------------------------------
AVERAGE_RATING
--------------
Classic Black Case
            5

Heavy Duty Case
            5

Clear Shock Case
           4.5


PRODUCT_NAME
--------------------------------------------------------------------------
AVERAGE_RATING
--------------
Crystal Clear Case
           4.5


10 rows selected.
```

- **Products requiring improvement:**

```
SQL> SELECT P.Product_Name,
  2         ROUND(AVG(R.Rating), 2) AS Average_Rating
  3  FROM Product P
  4  JOIN Review R
  5  ON P.Product_ID = R.Product_ID
  6  GROUP BY P.Product_ID, P.Product_Name
  7  HAVING AVG(R.Rating) < 3;

no rows selected
```

Report 3: Rating Distribution

The Rating Distribution report shows how ratings are distributed across products. It includes the number of 5-star ratings, the number of 4-star ratings, lower-rated products, and the overall distribution of ratings from 1 to 5.

This report provides a clear picture of customer satisfaction and helps identify products that need attention**.**

- **Number of 5-star ratings**

```
SQL> SELECT COUNT(*) AS Five_Star_Ratings
  2  FROM Review
  3  WHERE Rating = 5;

FIVE_STAR_RATINGS
-----------------
                8
```

- **Number of 4-star ratings**

```
SQL> SELECT COUNT(*) AS Four_Star_Ratings
  2  FROM Review
  3  WHERE Rating = 4;

FOUR_STAR_RATINGS
-----------------
                6
```

- **Number of low-rated reviews**

```
SQL> SELECT COUNT(*) AS Low_Rated_Reviews
  2  FROM Review
  3  WHERE Rating <= 2;

LOW_RATED_REVIEWS
-----------------
                0
```

- **Complete rating distribution**

```
SQL> SELECT Rating,
  2         COUNT(*) AS Number_Of_Ratings
  3  FROM Review
  4  GROUP BY Rating
  5  ORDER BY Rating DESC;

    RATING NUMBER_OF_RATINGS
---------- -----------------
         5                 8
         4                 6
```